Chapter 1 Conclusions and Outlook

In this thesis, we have developed and implemented a computational framework for the multi-orbital ladder-DΓA, enabling the study of two-particle correlation functions, single-particle quantities and superconducting transition temperatures in strongly correlated electron systems. A central focus was placed on the efficient treatment of vertex asymptotics, which significantly reduces numerical complexity for large frequency boxes and enables calculations down to very low temperatures. Combined with an optimized self-consistency cycle, where two-particle quantities feed back into single-particle properties via the Bethe-Salpeter, Schwinger-Dyson and Dyson equations, the implementation provides a robust and scalable approach to capture both local and non-local correlations beyond dynamical mean-field theory. The code is written in Python, exploits NumPy-based optimizations and uses mpi4py for distributed-memory parallelization, enabling large-scale multi-orbital simulations.

As with most scientific software, there are several promising directions for improving and extending this work:

First, while the implementation of the multi-orbital Eliashberg equation has been completed, its functionality has not yet been fully tested. A thorough validation of this routine would be an important next step, as it would allow for a direct and reliable treatment of superconducting instabilities and frequency-dependent pairing interactions. This, in turn, would open the door to a detailed investigation of the mechanisms driving superconductivity in strongly correlated systems with pronounced multi-orbital character.

Second, to make the framework applicable to modern research, it is essential to perform realistic multi-orbital calculations. By interfacing with ab-initio methods and incorporating material-specific input (e.g. from DFT + Wannier projections and a subsequent DMFT calculation), the code can be applied to experimentally relevant compounds such as cuprates, nickelates and iron-based superconductors.

Overall, the developed framework provides a solid foundation for investigating correlation effects and unconventional superconductivity in multi-orbital systems. By combining vertex asymptotics, an efficient self-consistency cycle and a scalable implementation, this thesis establishes a flexible basis for future research on strongly correlated materials.